

Meat Quality Assessment and Classification

CAPSTONE PROJECT PHASE-1

Phase – I Report

Submitted by Group 17

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This project report (Phase II) is submitted for the Project Viva-Voce examination held on

Supervisor

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INTRODUCTION

Food is one of the basic needs for humans derived from animal and vegetable foods. Healthy and clean foods will reflect healthy body condition and resistance to disease. Conversely, unhealthy foods will reduce the physical condition so susceptible to various diseases.

The main cause of unhealthy foods is the presence of congenital diseases. Foodborne diseases are a condition in which foods are contaminated by bacteria or other harmful substances that lead to a decrease in the quality of the food. In the USA, about 76 million food-borne illness cases occur each year.

[1] The economic losses caused by diseases from the major pathogen are estimated at more than \$35 billion for medical costs and lost productivity annually.

[2] The European Food Safety Authority has reported that more than 15% of the populations in Europe suffer from foodborne diseases. In Indonesia, there were 1068 cases of food poisoning in 2016.

[3] India is ranked 8th in global ranking of total food poisoning abroad occurring in the country as per Annie Daly an editor of yahoo life. Meat is one example of widely consumed foodstuff but it is also a kind of perishable food. The Food and Agriculture Organization (FAO) of the United Nations estimates that beef has a high demand and will still become one of the main sources of animal-based protein (ABP) in the world until 2050. The report shows that the total consumption will increase both in developed countries and developing countries.

[4].A chemical based assessment of meat leads to destruction of the sample under test and the sample cannot be used further. However, with image processing, only an image is used for analysis.

This is achieved by using an image processing technique, which is a non destructive and non-hazardous method of assessment, based on photography and analysis of its color variations and evaluating its freshness with the help of a machine learning model.

1.1 Motivation

The global meat industry faces significant challenges in maintaining consistency in quality assessment due to reliance on subjective human judgment. Furthermore, the growing consumer demand for transparency and quality assurance has amplified the need for standardized solutions. Existing processes often lack scalability and depend heavily on skilled labor, making them inefficient for large-scale operations. AI and ML present an opportunity to address these issues through automation, improving speed, reliability, and cost-efficiency in quality gradation.

By integrating advanced image processing and machine learning techniques, this project aims to streamline meat grading while enhancing precision, enabling producers and distributors to meet regulatory requirements and consumer expectations more effectively.

1.2 Objective

The primary goal of this project is to develop an AI-driven system for automated meat quality assessment. The specific objectives include:

1. **Developing a Robust Classification Model:** Utilize deep learning techniques, particularly convolutional neural networks (CNNs), to classify meat quality based on key visual features like marbling, texture, and color.
2. **Optimizing Model Accuracy:** Apply advanced ML techniques, including hyperparameter tuning and augmentation, to achieve high accuracy and generalizability across diverse datasets.
3. **Creating a User-Friendly Interface:** Design an intuitive, accessible platform that allows non-technical users to upload images and retrieve real-time grading results.
4. **Scalability and Deployment:** Ensure the system can handle large-scale datasets and integrate seamlessly into existing industrial workflows.
5. **Long-Term Adaptability:** Enable the system to adapt to new data, allowing continuous improvement and customization to specific industry needs.

EXISTING WORK

1. Gradation using KNN

Trientin[5] proposed the classification of beef's freshness through the sensory analysis of the samples' color, using the K-nearest neighbors (KNN) model. The authors captured the samples' images in a controlled environment, using a digital camera. RGB parameters were extracted and converted to HSV parameters to check the brightness difference. The overall accuracy of the model using KNN stands at 75%.

Taheri-Garavand [6] used a method based on the KNN to assess the common carp's freshness (*Cyprinus carpio*) during storage on ice. Sample images were captured in a controlled environment. Parameters of the RGB, his, and L*a*b* color spaces were extracted from 1344 images of samples. The overall accuracy of the model using KNN stands to 90.48%.

S Agustin [7] exercised Beef Image Classification using K-Nearest Neighbor Algorithm for Identification Quality and Freshness. The authors converted the RGB to Binary image. A thresholding process was used to separate the pixels based on the gray level. Image segmentation/edge detection is done to increase the appearance between the boundary line of an area or object in the image. Then the Gray Level Co-Occurrence Matrix (GLCM) of the image is found. The K-Nearest Neighbor is used to classify objects, based on learning data that is close to the object, according to the number of their closest neighbors or k values. The proximity or distance of the neighbor is usually calculated based on Euclidean distance. The results showed that the performance of the system using the KNN method to identify the quality of meat based on color and texture can detect the type of beef and the amount of accuracy of 91.0667%.

Christell Faith D. Lumogdang[8] applied a studio-type chamber for capturing images and detecting gases. The box is made up of glass with two fixed Light Emitting Diodes (LED) on both sides for proper lighting and ray distribution. Found in the center of the chamber ceiling is the mounted Raspberry Pi Camera which captures the sample pork meat placed on the raised flooring or platform below it. On the left wall of the case are the gas sensors, MQ- 135 and MQ-136. Beside the studio-type chamber is the circuitry box, which rooms the Arduino Uno, Raspberry Pi 3, exhaust fan, and the 7-inch LCD monitor. The overall accuracy of the model using KNN stands to 93.33%.

Kenan Lugatiman [9] employed an app which used computer vision for RGB extraction and k-Nearest Neighbors (k-NN) algorithm for classification and Waikato Environment for Knowledge Analysis (WEKA) in terms of the number of hours from slaughter. The overall accuracy of the model using KNN stands to 86.76%.

2. Gradation using Linear Regression

Sun, Young, Liu, Chen, and Newman [10] investigated pork's freshness through its color characteristics, observed in digital images. The study compared the performance of the traditional regression methods. Eighteen image color features were extracted from three different RGB (red, green, blue) models, HSI (hue, saturation, intensity), and L*a*b* color spaces. Two comparable regression models (linear and stepwise) were used to evaluate prediction results of pork color at- Tributes. The proposed linear regression model had an accuracy of 83%.

3. Gradation using Logistic Regression

Nachiketa Hebbar [11] utilized a sensor that measures the concentration of oxygen and ammonia levels in a food and a machine learning model using logistic regression is used to predict if the given food item is spoiled or not. It can also be deployed again to predict shelf life. The proposed linear regression model had an accuracy of 100%.

4. Gradation using SVM

Xiao Guan [12] suggest the following method - The steps of freshness assessment of meat samples based on QPSO-SVM are described as follows:

Step 1: Initialize the original data by normalization and then form a training sample set.

Step 2: Based on the RBF kernel function, call the embedded QPSO algorithm and get the optimal parameters. Construct the QP problem (equation 1) of SVM.

Step 3: Solve the optimization problem (equation 2) and compute the classification result according to equation 6.

The proposed SVM model had an accuracy of 92.8%.

Arsalane [13] presented the implementation of the principal component analysis (PCA) and support vector machine (SVM) models to classify and predict the freshness of beef. A data set of eighty-one beef images was analyzed based on the HSI color space. The beef images were captured in a controlled environment. The authors used the PCA model as a projection model and the SVM to classify and identify beef. The results obtained from the PCA projection model show the projection of three groups representing the freshness of beef meat during the days of refrigerated storage. The SVM model got a 100% success rate of classification and identification.

Taheri-Garavand[14] employed a method based on the SVM to assess the common carp's freshness (*Cyprinus carpio*) during storage on ice. Sample images were captured in a controlled environment. Parameters of the RGB, HSI, and $L^*a^*b^*$ color spaces were extracted from 1344 images of samples. The overall accuracy of the model using SVM stands to 91.52%.

5. Gradation using K-Means

Malay Kishore Dutta [15] made an Image processing based method to assess fish quality and freshness. The RGB image is converted to Lab color space model. The Lab color space model is designed to approximate human vision and it suits the requirement to segment the gills from the fish image. K-means clustering algorithm works in 2 steps: Assignment phase and Update phase. Three clusters are formed. Feature extraction is done from the red channel in the wavelet transformation domain using. First, second and third level decomposition is performed. The statistical features of coefficients (mean and standard deviation) obtained at each level. The features extracted from horizontal coefficients at level 3 for all days of observation are analyzed for the variation pattern. The entire framework is used to divide the images into three freshness ranges. The accuracy obtained using this model is 95.833%.

Winiarti [16] harnessed identifying beef quality by sensory analysis of the color observed in samples photographed by a digital camera. The proposed system captures the sample image and calculates its' RGB color space parameters. The authors used the histogram for each color channel in the sample to group 40 meat samples into four clusters, using the K-means model representing four categories: very viable, viable, less viable, or unfeasible. The determination of the categories is obtained based on the calculation of the Euclidean distance. The system classified forty meat samples, demonstrating that color parameters can group meat samples into different clusters. The accuracy obtained using this model is not given.

SUMMARY -

Classifier	Author	Type of sample	Number of sample	Color space	Accuracy (%)
K Nearest Neighbors	Trientin, Hidayat, and Darana	BEEF	Uninformed	RGB,HSB	75
	Taheri-Garavand, Fatahi, Banan, and Makino	CARP	1344	RGB, L*a*b*	90.48
	S Agustin, R Dijaya	BEEF	60	RGB, HSI, Grayscale, Binary	91.0667
	Christell Faith D. Lumogdang, Marianne G. Wata, Christell Faith D. Lumogdang, Stephone Jone S. Loyola, Randy E. Angelia, Hanna Leah P. Angelia	PORK	75	RGB, Binary Gradient	93.33
	Kenan Lugatiman, Crisanto Fabiana, Jairo Echavia, Jetron J. Adtoon	TUNA	90	RGB, Binary Gradient	86.67

K Means	Malay Kishore Dutta , Ashish Issac , Navroj Minhas, Biplab Sarkar	FISH	24	RGB, XYZ, Lab color space	95.833
	Jae Moon Lee,In Hwan Jung, Kitae Hwang	BEEF	300	Uninformed	75
Linear Regression	Sun, Young, Liu, Chen, and Newman	PORK	Uninformed	RGB, HSI, L*a*b*	83
Logistic regression	Nachiketa Hebbar	MEAT FRUIT FISH VEGETABLE	40	Uninformed	100
Support Vector Machine(SVM)	Xiao Guan, Jing Liu, Qingrong Huang, And Jingjun Li	PORK BEEF MUTTON SHRIMP	112(28*4)	Uninformed	92.8
	Arsalane, Barbri, Tabyaoui, Klilou, Rhofir, and Halimi	BEEF	Uninformed	HSI	100
	Taheri-Garavand, Fatahi, Banan, and Makino	CARP	1344	RGB, HSI, L*a*b*	91.52

DATASET

1. Fish - The live Fish (Pabda) were sampled live from a local aquatic products market in Kolkata of India and kept in three hundred liters aquariums for 24 h. The average weight and average length of fishes were 90.40 ± 1.20 g and 21.60 ± 0.50 cm respectively. The pond water from which it was collected was free from any pathogenic infestation and toxic residues.

Fishes from the aquariums were taken out and placed into chilled water for sudden death to avoid rigor mortis. The fishes thereafter were preserved for imaging study in thermocol boxes ($28 \times 18 \times 12$ cm³) with a fish to ice ratio of 1:2. Images of fish were taken using a digital camera and the distance between the fish and the camera was as far as 10 cm. The captured images are of the size 601 x 361 pixels.

The fish images were captured starting from day one of death and at every two days' interval till the 10th day. Fish was stored in a freezer in the intermediate days with a temperature of 0 degree Celsius. Fish images classified as consumable (Refer Fig 1.1.1) were taken till 4-5 days from day one of death, and fish classified as non-consumable (Refer Fig 1.1.2) were taken from 5-6 days of death till the 10th day. The process of taking the image of fish meat is carried out in an open area which is illuminated by the natural sunlight. After the process of taking the image data, a certain part of the image object would be cropped[4].

The proposed image processing based method of freshness identification in fish samples involves feature extraction from the color difference in stomach area. To extract the accurate and discriminatory features from the image, the portion of the image which contains maximum information is required to be segmented from the whole image. After the process of taking the image data, the RGB image (Refer Fig 1.1.1 and 1.1.2) is then converted into HSV [1] (Refer Fig 1.1.3 and 1.1.4) for further processing.



Fig. 1.1.1 Consumable (RGB)



Fig. 1.1.2 Non-consumable (RGB)

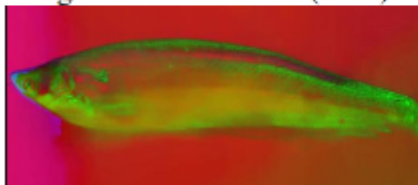


Fig. 1.1.3 Consumable (HSV)

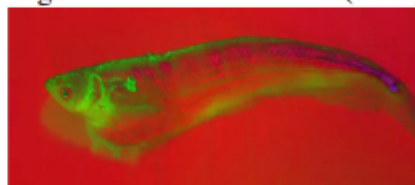


Fig. 1.1.4 Non-Consumable (HSV)

2. Chicken - The chicken meat was sampled from a local market where the chicken was brought alive from a nearby poultry. Breast meat portion was used as a sample for the dataset. The chicken breast is cut to various lengths and widths but with almost uniform thickness, approximately 0.5 cm.

The process of capturing chicken meat image data used a digital camera and the distance between the chicken meat and the camera was as far as 10 cm. The captured images are of the size 601 x 361 pixels. The chicken meat images were captured starting from day one of slaughter and at every two days' interval till 13th day. Meat was stored in a freezer in the intermediate days with a temperature of 0 degree Celsius. Meat images classified as consumable (Refer Fig 1.2.1) were taken till 5-6 days from day one of death, and meat classified as non-consumable (Refer Fig 1.2.2) were taken from 5-6 days of death till the 13th day. The process of taking the image of chicken meat is carried out in an open area which is illuminated by the natural sunlight. After the process of taking the image data, a certain part of the image object would be cropped.

The proposed image processing based method of freshness identification in chicken meat samples involves feature extraction from the color difference. [3] To extract the accurate and discriminatory features from the image, the portion of the image which contains maximum information is required to be segmented from the whole image. After the process of taking the image data, the RGB image (Refer Fig 1.2.1 and 1.2.2) is then converted into HSV [1] (Refer Fig 1.2.3 and 1.2.4) for further processing.

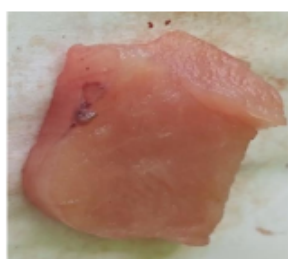


Fig. 1.2.1 Consumable (RGB)



Fig. 1.2.2 Non-Consumable (RGB)

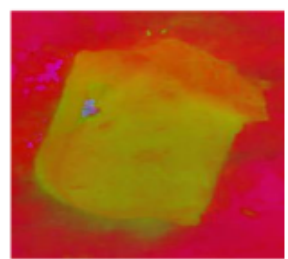


Fig. 1.2.3 Consumable (HSV)

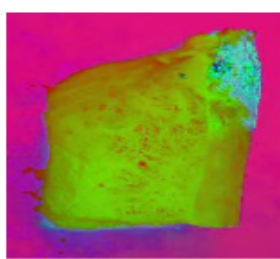


Fig. 1.2.4 Non-Consumable (HSV)

3. Prawn - Fresh white-leg prawn each of approximately 24 ± 2 g in weight and 17 ± 2 cm in length were sampled from a local aquatic products market Kolkata of India. The shrimp were kept in seawater with ice and were transferred to the laboratory within 2 hours of purchase. After prawn with signs of visual defect or breakage were removed, the remaining shrimp were picked out. The shrimp were washed clean with tap water. The meat sample was collected immediately by hand with gloves to prevent contamination.

The prawn images were captured starting from day one of purchase and at every one day interval till 7th day. It was stored in a freezer in the intermediate days with a temperature of 0 degree Celsius. Images classified as consumable (Refer Fig 1.3.1) were taken till 3-4 days from the day one of death, and images classified as non-consumable (Refer Fig 1.3.2) were taken from 3-4 days of death till the 7th day. The process of taking the image of prawn is carried out in an open area which is illuminated by the natural sunlight. After the process of taking the image data, a certain part of the image object would be cropped.

The proposed image processing based method of freshness identification in prawn samples involves feature extraction from the color difference. To extract the accurate and discriminatory features from the image, the portion of the image which contains maximum information is required to be segmented from the whole image. After the process of taking the image data, the RGB image (Refer Fig 1.3.1 and 1.3.2) is then converted into HSV [1] (Refer Fig 1.3.3 and 1.3.4) images for further processing



Fig. 1.3.1 Consumable (RGB)



Fig. 1.3.2 Non-Consumable (RGB)

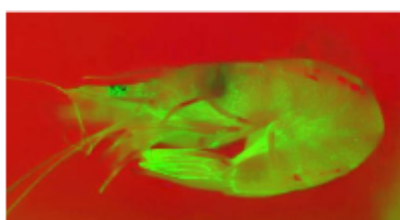


Fig. 1.3.3 Consumable (HSV)

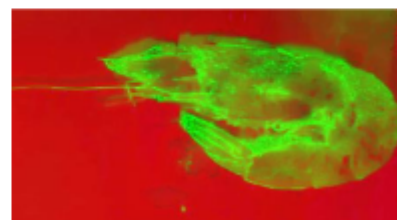


Fig. 1.3.4 Non-Consumable (HSV)

METHODOLOGY

1. Classification using Naive Bayes

Naive Bayes classifiers are a family of simple "probabilistic classifiers" based on applying Bayes' theorem with strong (naive) independence assumptions between the features (see Bayes classifier). They are among the simplest Bayesian network models, but coupled with kernel density estimation, they can achieve high accuracy levels.

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Using Bayes theorem, we can find the probability of **A** happening, given that **B** has occurred (Refer Fig 2.1.0). Here, **B** is the evidence and **A** is the hypothesis. The assumption made here is that the predictors/features are independent. That is, the presence of one particular feature does not affect the other. Hence it is called naive.

The proposed system uses two color spaces (RGB and HSV) to train the model for each of the three dataset[5]. After training the system the model classified each of the dataset into two classifications: consumable and non-consumable. The result of the test dataset for each of the dataset is shown in the figures below as a confusion matrix (Refer Fig. 2.1.1 to 2.1.6).

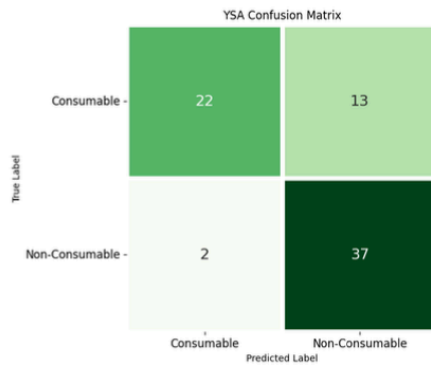


Fig. 2.1.1 Chicken Dataset (RGB)

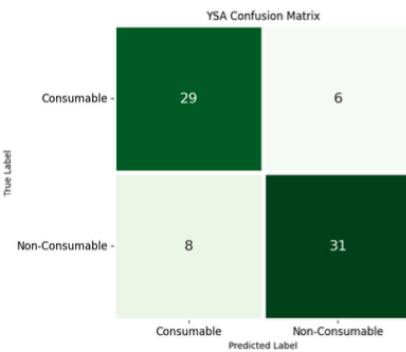


Fig. 2.1.2 Chicken Dataset (HSV)

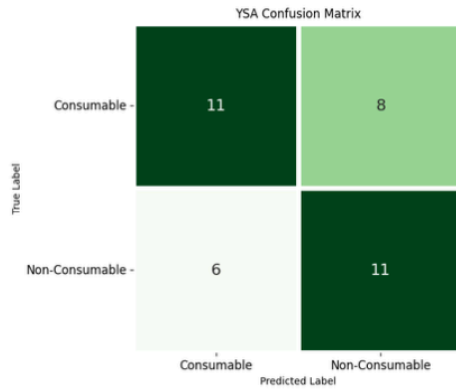


Fig. 2.1.3 Fish Dataset (RGB)

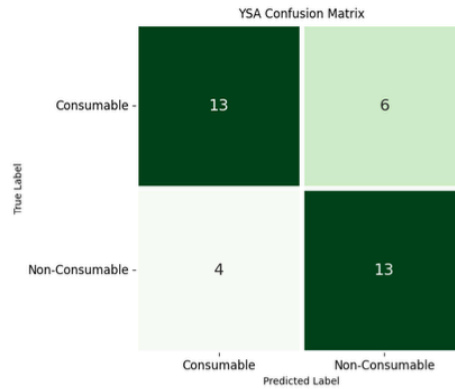


Fig. 2.1.4 Fish Dataset (HSV)

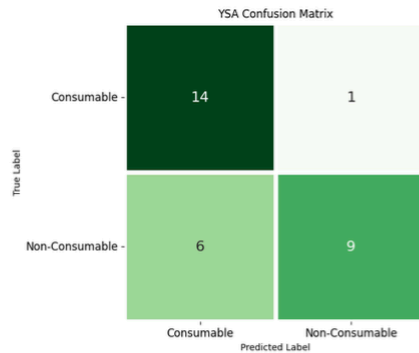


Fig. 2.1.5 Prawn Dataset (RGB)



Fig. 2.1.6 Prawn Dataset (HSV)

As we can see Chicken (Refer Fig. 2.1.1 and 2.1.2) and Prawn (Refer Fig. 2.1.5 and 2.1.6) provided a better accuracy than Fish (Refer Fig. 2.1.3 and 2.1.4) in this model. Chicken in HSV Color Space (Refer Fig. 2.1.2) provided the best accuracy under this model.

2. Classification using KNN

In statistics, the ***k*-nearest neighbors algorithm (*k*-NN)** is a non-parametric supervised learning method first developed by Evelyn Fix and Joseph Hodges in 1951, and later expanded by Thomas Cover. It is used for classification and regression. In both cases, the input consists of the *k* closest training examples in a data set[6]. The output depends on whether *k*-NN is used for classification or regression:

In *k*-NN classification, the output is a class membership. An object is classified by a plurality vote of its neighbors, with the object being assigned to the class most common among its *k* nearest neighbors (*k* is a positive integer, typically small). If *k* = 1, then the object is simply assigned to the class of that single nearest neighbor. In *k*-NN regression, the output is the property value for the object. This value is the average of the values of *k* nearest neighbors. *k*-NN is a type of classification where the function is only approximated locally and all computation is deferred until function evaluation. Since this algorithm

relies on distance for classification, if the features represent different physical units or come in vastly different scales then normalizing the training data can improve its accuracy dramatically.

Both for classification and regression, a useful technique can be to assign weights to the contributions of the neighbors, so that the nearer neighbors contribute more to the average than the more distant ones. For example, a common weighting scheme consists in giving each neighbor a weight of $1/d$, where d is the distance to the neighbor[6]. The proposed system uses two color spaces (RGB and HSV) to train the model for each of the three dataset. After training the system the model classified each of the dataset into two classifications: consumable and non-consumable. The result of the test dataset for each of the dataset is shown in the figures below as a confusion matrix (Refer Fig. 2.2.1 to 2.2.6).

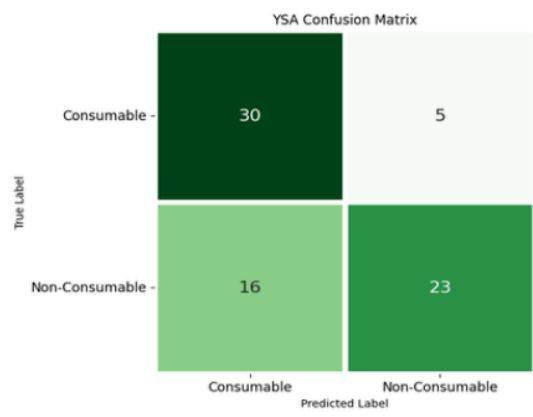


Fig. 2.2.1 Chicken Dataset (RGB)

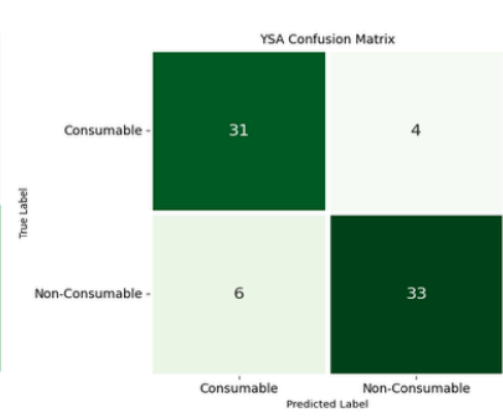


Fig. 2.2.2 Chicken Dataset (HSV)

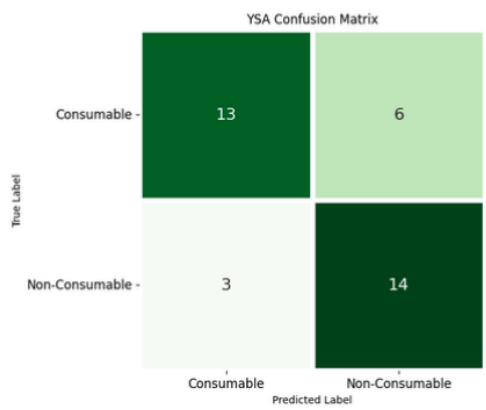


Fig. 2.2.3 Fish Dataset (RGB)

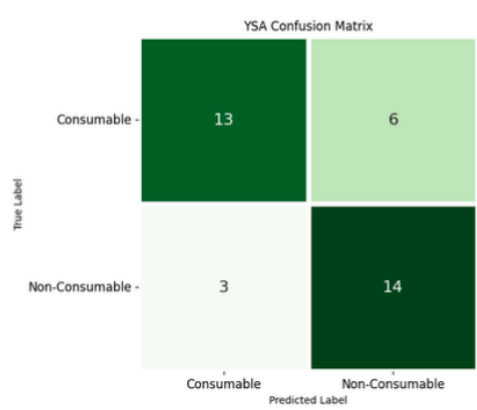


Fig. 2.2.4 Fish Dataset (HSV)

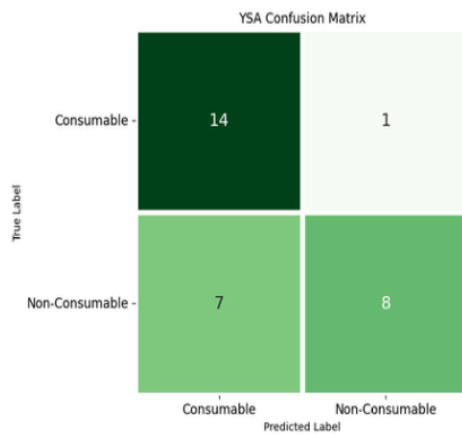


Fig. 2.2.5 Prawn Dataset (RGB)

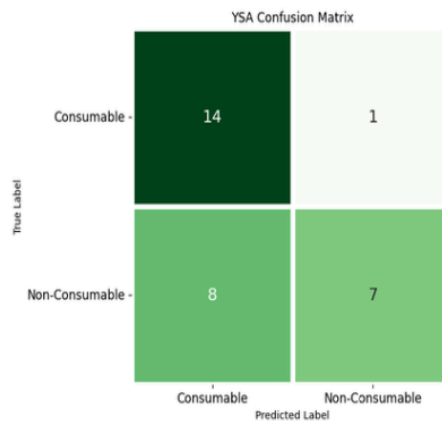


Fig. 2.2.6 Prawn Dataset (HSV)

As we can see all three datasets provide almost the same accuracy in RGB but in HSV color space Chicken (Refer Fig. 2.2.2) provides better results than the other two

3. Classification using SVM

In machine learning, **support vector machines (SVMs, also support vector networks)** are supervised learning models with associated learning algorithms that analyze data for classification and regression analysis. Developed at AT&T Bell Laboratories by Vladimir Vapnik with colleagues (Boser et al., 1992, Guyon et al., 1993, Cortes and Vapnik, 1995, Vapnik et al., 1997) SVMs are one of the most robust prediction methods, being based on statistical learning frameworks or VC theory proposed by Vapnik (1982, 1995) and Chervonenkis (1974)[7]. Given a set of training examples, each marked as belonging to one of two categories, an SVM training algorithm builds a model that assigns new examples to one category or the other, making it a non-probabilistic binary linear classifier (although methods such as Platt scaling exist to use SVM in a probabilistic classification setting). SVM maps training examples to points in space so as to maximize the width of the gap between the two categories[7]. New examples are then mapped into that same space and predicted to belong to a category based on which side of the gap they fall. The proposed system uses two color spaces (RGB and HSV) to train the model for each of the three dataset. After training the system the model classified each of the dataset into two classifications : consumable and non-consumable. The result of the test dataset for each of the dataset is shown in the figures below as a confusion matrix (Refer Fig. 2.3.1 to 2.3.6).

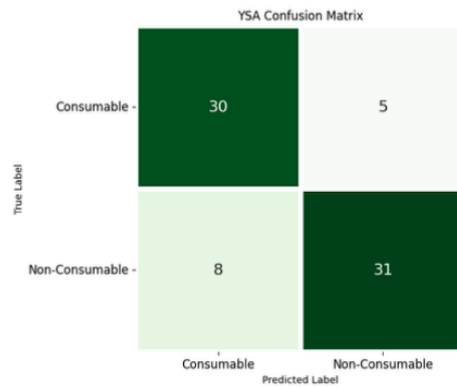


Fig. 2.3.1 Chicken Dataset (RGB)

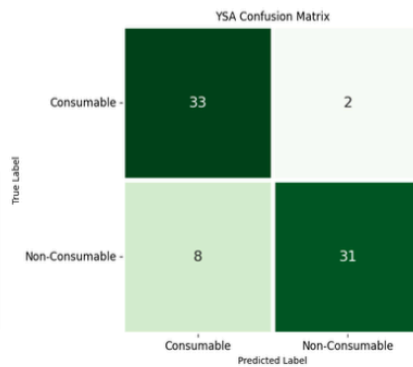


Fig. 2.3.2 Chicken Dataset (HSV)

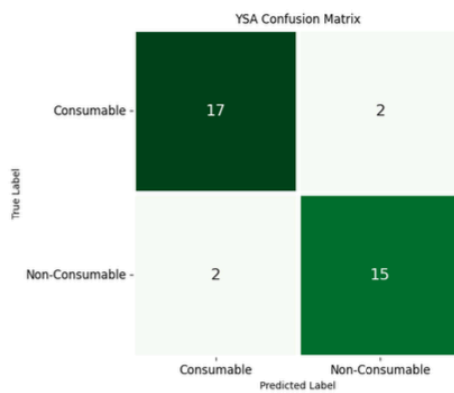


Fig. 2.3.3 Fish Dataset (RGB)

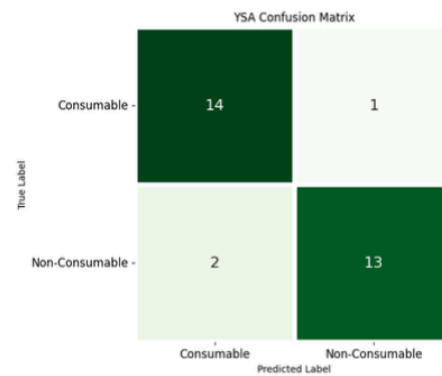


Fig. 2.3.4 Fish Dataset (HSV)

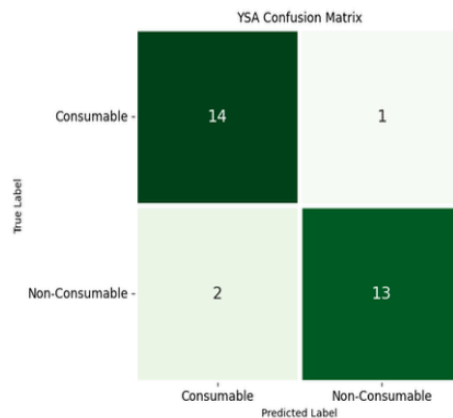


Fig. 2.3.5 Prawn Dataset (RGB)

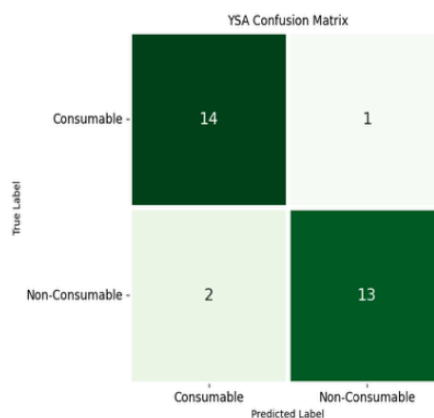


Fig. 2.3.6 Prawn Dataset (HSV)

As we can see the datasets containing prawn provided excellent accuracy under this model in both color spaces (Refer Fig. 2.3.5 and 2.3.6). The other two datasets also provided good output in both color spaces.

4. Classification using Random Forest

Random forests or random decision forests is an ensemble learning method for classification, regression and other tasks that operates by constructing a multitude of decision trees at training time[8]. For classification tasks, the output of the random forest is the class selected by most trees. For regression tasks, the mean or average prediction of the individual trees is returned. Random decision forests correct for decision trees' habit of overfitting to their training set. Random forests generally outperform decision trees, but their accuracy is lower than gradient boosted trees.[citation needed] However, data characteristics can affect their performance[8]. The proposed system uses two color spaces (RGB and HSV) to train the model for each of the three dataset. After training the system the model classified each of the dataset into two classifications : consumable and non-consumable. The result of the test dataset for each of the dataset is shown in the figures below as a confusion matrix (Refer Fig. 2.4.1 to 2.4.6).

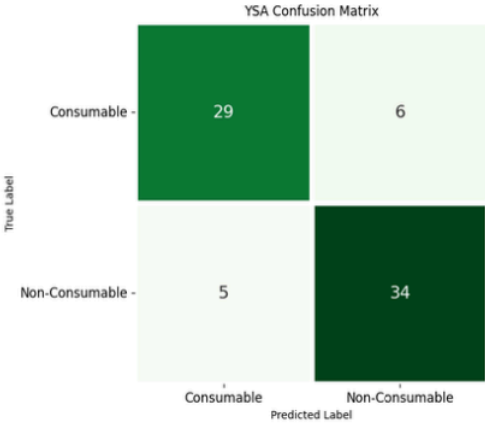


Fig. 2.4.1 Chicken Dataset (RGB)

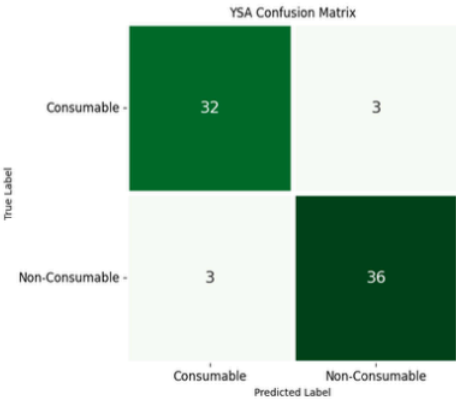


Fig. 2.4.2 Chicken Dataset (HSV)

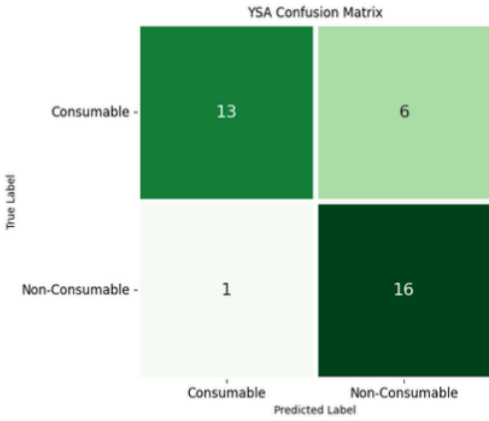


Fig. 2.4.3 Fish Dataset (RGB)

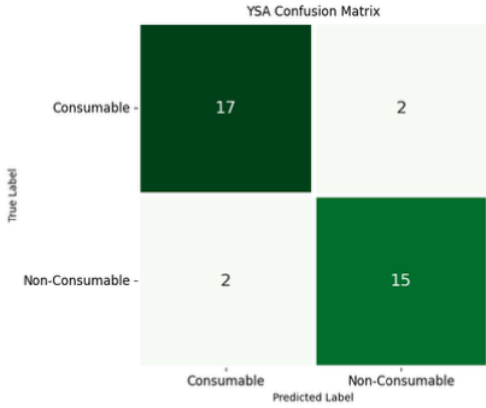


Fig. 2.4.4 Fish Dataset (HSV)

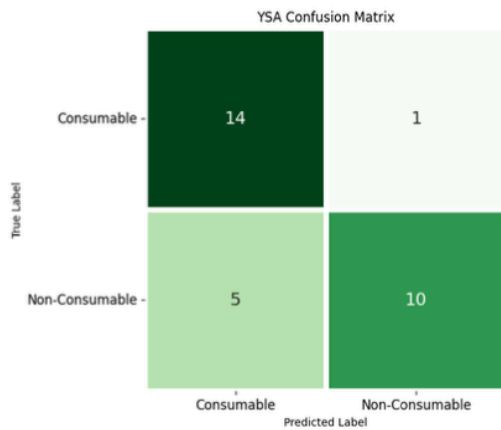


Fig. 2.4.5 Prawn Dataset (RGB)

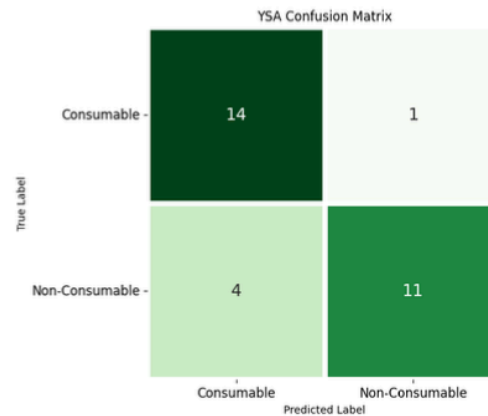


Fig. 2.4.6 Prawn Dataset (HSV)

As we can see Chicken and Fish equally provided good results under this method mainly in HSV color spaces (Refer Fig. 2.4.2 and 2.4.4). Prawn gave somewhat intermediate accuracy in HSV Color space (Refer Fig 2.4.6).

RESULTS

The experiment was to predict the freshness of meat on a given color space, and the results are the accuracies on the different dataset. The results show that image classification can be used to classify good (consumable) and bad (non-consumable) meat. One of the important results was quality of image and color space played a vital role in predicting the quality of meat. The results classified meat according to their freshness criteria. The biggest strength of the study was the model being simplistic, but limitations are less accurate compared to deep learning models, which are very complex. The variables were like, Random Forest being the best model for chicken classifier whereas for prawn the best was SVM.

The complete analysis is shown in table (Refer Table 1) below.

Table 1: Accuracy observed for different models for different datasets under different color spaces.

Classifier	Type of sample	Number of sample	Color space	Accuracy * (%)
Naive Bayes	CHICKEN	310	RGB	79.729
			HSV	89.081
	FISH	140	RGB	61.111
			HSV	72.222
	PRAWN	122	RGB	76.666
			HSV	83.333
KNN	CHICKEN	310	RGB	71.621
			HSV	86.486
	FISH	140	RGB	72.012
			HSV	75.023
	PRAWN	122	RGB	73.333

			HSV	70
SVM	CHICKEN	310	RGB	82.432
			HSV	86.486
	FISH	140	RGB	88.888
			HSV	88.888
	PRAWN	122	RGB	90
			HSV	90
Random Forest	CHICKEN	310	RGB	85.135
			HSV	91.891
	FISH	140	RGB	80.555
			HSV	88.888
	PRAWN	122	RGB	80
			HSV	83.333

* Taken by average from multiple results by randomizing the dataset.

When working with Chicken dataset in both the color spaces, we have observed that in all the algorithms HSV color space provided much better results than its RGB counterpart (Refer Fig 3.1). Out of all the algorithms Random Forest showed the best accuracy in both color spaces.

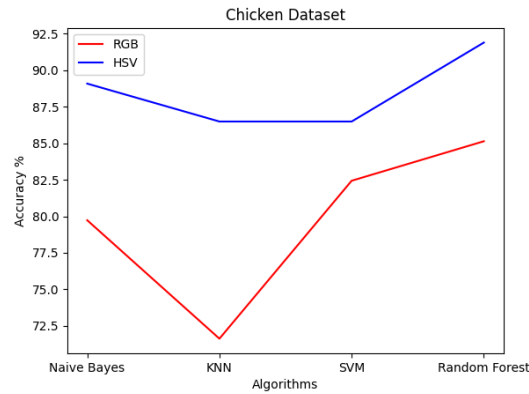


Fig. 3.1 Chicken dataset

When working with Fish dataset in both the color spaces, we have observed that in all the algorithms HSV color space provided slightly better results than its RGB counterpart (Refer Fig 3.1). Out of all the algorithms Random Forest and SVM showed the best accuracy in HSV color space.

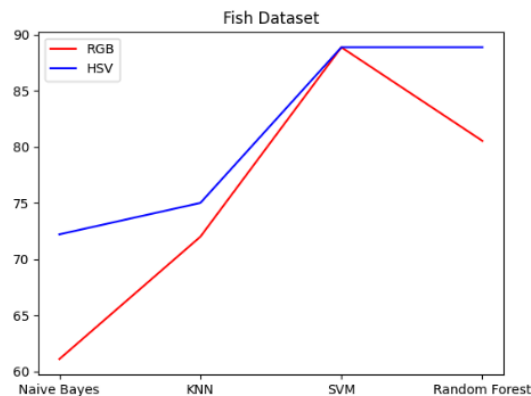


Fig. 3.2 Fish dataset

Fig. 3.2 Fish dataset

When working with Prawn dataset in both the color spaces, we have observed that in most of the algorithms in HSV and RGB color spaces provided almost the same accuracy (Refer Fig 3.1). Out of all the algorithms SVM showed the best accuracy in both color spaces.

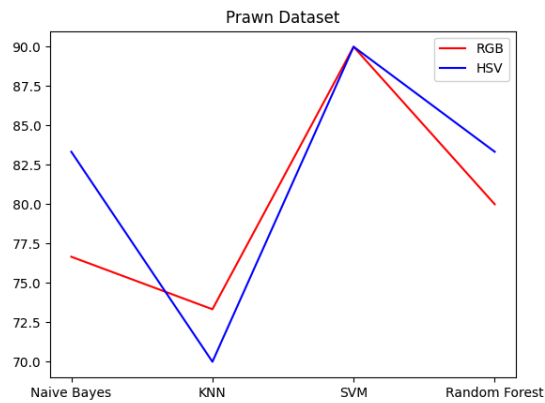


Fig 3.3 Prawn dataset

When working with Prawn dataset in both the color spaces, we have observed that in most of the algorithms in HSV and RGB color spaces provided almost the same accuracy (Refer Fig 3.1). Out of all the algorithms SVM showed the best accuracy in both color spaces.

Individual Contribution by Members -

Member 1 Name - Divyansh Bhilware

Member 1 Registration No - 21BCE10932

Contribution -

Divyansh worked alongside Harshit in the research and implementation of machine learning models. He focused on optimizing the hyperparameters and ensuring the models' scalability and efficiency. His tasks included implementing cross-validation techniques to prevent overfitting and running experiments to compare the performance of different algorithms. He also contributed to integrating pre-trained models where applicable and improving the computational efficiency of the pipelines. Divyansh's systematic approach to troubleshooting and model refinement greatly enhanced the overall performance of the project.

Member 2 Name - Priyanshu Singh

Member 2 Registration No - 21BCE10942

Contribution - Priyanshu was responsible for testing the final machine learning model for meat quality detection and assessment. He meticulously evaluated the model's performance using various testing datasets and metrics such as accuracy, precision, recall, and F1-score. His role included ensuring that the model was robust, reliable, and capable of handling edge cases effectively. Additionally, Priyanshu prepared a comprehensive project report and presentation to summarize the team's efforts. He structured the documentation to include the project's motivation, methodology, results, and future scope. His contribution ensured clarity and professionalism in the deliverables, making the project presentation ready for stakeholders.

Member 3 Name - Rohan Roy

Member 3 Registration No - 21BCE11181

Contribution - **Quality Assurance/Tester:**

Rohan was pivotal in the preparation and preprocessing of the dataset used for training the machine learning models. His work involved collecting relevant data, cleaning it by handling missing or noisy values, and ensuring the data was representative of real-world meat quality variations. He implemented data augmentation techniques to expand the dataset, thereby improving the robustness of the models. Rohan was also involved in training the models and monitoring their performance during iterative training sessions. His attention to detail in dataset preparation ensured that the models achieved high accuracy and reliability.

Member 4 Name - Harshit Srivastava

Member 4 Registration No - 21BCE11325

Contribution - Harshit contributed significantly to the research and implementation of the ML models. He analyzed various machine learning algorithms suitable for the problem, such as support vector machines, random forests, and convolutional neural networks (CNNs). Harshit designed the initial architecture for the chosen model, emphasizing efficient feature extraction and prediction accuracy. He also implemented these models in Python using frameworks like TensorFlow or PyTorch and ensured seamless integration with the datasets prepared by the team. Harshit's expertise in coding and algorithmic implementation was vital in establishing the foundation for the project's success.

Member 5 Name - Arghya Pal

Member 5 Registration No - 21BCE11336

Contribution -

Arghya collaborated with Rohan on dataset preparation and model training. He ensured the datasets were formatted correctly and split appropriately into training, validation, and testing sets. Arghya implemented feature engineering techniques to extract meaningful patterns from the data and contributed to building pipelines for seamless data processing. During model training, he worked on setting up and fine-tuning the training configurations, including batch sizes, learning rates, and epochs. Arghya's proactive involvement in handling technical challenges during training was critical for the timely completion of the project.

CONCLUSION

Machine learning models used for the above problem for each of the dataset some particular models provided better accuracy than the others. For example, Chicken dataset in HSV Color Space and in Random Forest model provided the maximum accuracy which is over 90% whereas Prawn and Fish in both RGB and HSV color space provided maximum accuracy of over 85% on SVM Model.

So, our aim will be to optimize these particular models for those datasets to provide better results for far more wide instances of those datasets. Addition to that we will try more complex deep learning models like Convolutional Neural Network and compare the results with the above machine learning models and choose accordingly for further experimentation, keeping in mind the computational complexity and the platform in which the project will be implemented.

Accurate meat freshness assessment is crucial for the problem of food quality. In general, meat freshness cannot be assessed accurately by any single conventional index because every index reflects only partial characteristics of a meat sample. To conclude, artificial vision and machine learning is a reliable technique, and it has shown its efficiency in many applications related to meat assessment.

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